HACKATHON PITCH Q&A GUIDE

Green Logistics: 10 Key Questions & Answers

Core insights, challenge analysis, and technological solutions explained in a single question-and-answer format per slide.

QUESTION 01

Q: What is the main environmental problem with global freight logistics today?

Answer: Logistics generates 11% of global greenhouse emissions, largely driven by inefficient transport routes, low capacity utilization, and reliance on fossil-fuel heavy trucks.

QUESTION 02

Q: Why is last-mile urban delivery particularly harmful to cities?

Answer: Last-mile fulfillment causes 35% of urban traffic congestion, leading to severe idle tailpipe emissions, delayed deliveries, and dangerous street air pollution.

QUESTION 03

Q: What core solution does this hackathon project propose?

Answer: An integrated platform offering AI carbon-aware routing, collaborative peer-to-peer load sharing, and automated eco-packaging volume tracking.

QUESTION 04

Q: How does AI routing reduce emissions compared to standard GPS?

Answer: It prioritizes total carbon reduction over distance by calculating road elevation resistance, avoiding idle traffic, and syncing EV charging stops with renewable grid availability.

QUESTION 05

Q: How does the collaborative network eliminate empty return trips?

Answer: It matches shippers with empty truck bed capacity on backhaul routes in real-time, reducing wasted mileage by up to 28% without exposing carrier sensitive data.

QUESTION 06

Q: What technologies power the backend optimization engine?

Answer: A FastAPI asynchronous backend combined with Google OR-Tools constraint solvers to perform complex vehicle routing calculations in real-time.

QUESTION 07

Q: Why will logistics carriers adopt this platform financially?

Answer: Operating empty trucks wastes fuel and labor; monetizing backhaul capacity cuts operating costs by 30% while fulfilling corporate ESG compliance goals.

QUESTION 08

Q: What overall emissions impact can deployment achieve?

Answer: Deployment across regional delivery networks projects a 42% reduction in CO2 emissions per route using verified GLEC carbon accounting standards.

QUESTION 09

Q: What is the phased plan to bring this prototype to market?

Answer: Start with campus pilot route testing, partner with regional courier fleets for live beta trials, and scale to full municipal logistics integration.

QUESTION 10

Q: Why should this solution win the hackathon?

Answer: It delivers a practical, technology-driven solution combining immediate financial savings for carriers with massive, measurable decarbonization for our planet.